Attention Filtering Network Based on Branch Transformer for Change Detection in Remote Sensing Images

Yu Shangguan, Jinjiang Li, Yepeng Liu, Fan Zhang, and Caiming Zhang

***Abstract*— The emergence of high-resolution (HR) remote sensing imagery showcases the continual advancements in remote sensing technology but also sets higher demands for related tasks in the field, including remote sensing image change detection. Due to their outstanding performance in extracting salient features, convolutional neural networks (CNNs) have played a significant role and become widely utilized in many computer vision tasks. The encoder–decoder structure has confirmed the effectiveness of integrating multilevel feature information, as it allows for the synthesis of both local and global information of features. The exploration of the potential relationships between multilevel features and their efficient integration remains of significant importance. Furthermore, thanks to the advent of the transformer, many modern approaches have seen great improvements in high-level semantic understanding of images. In this article, we propose an attention-filtering network based on a branch transformer for effective change detection in remote sensing images. A hybrid attention fusion module (HAFM) is used to efficiently fuse features of different granularities and perform progressive information filtering on the extracted multilevel features to obtain an effective change feature. We also propose a branch transformer block (BTB) to efficiently aggregate global long-range dependencies and spatial details from the change feature. Extensive comparative experiments conducted on three different HR remote sensing datasets have verified the effectiveness of our method.**

***Index Terms*— Branch transformer block (BTB), change detection, convolutional neural network (CNN), high-resolution (HR) remote sensing image, hybrid attention fusion module (HAFM).**

I. Introduction

CHANGE detection (CD) task applied to remote sensing images needs to perceive and recognize the changes of ground objects in paired satellite images from different times and the same region [1]. As shown in Fig. 1, for two remote sensing images reflecting regional information at different times, CD technology needs to compare the changes over time between the two to obtain a binary mask that can indicate significant changes. CD technology can provide precise and detailed information on the changes in surface targets, which can help us obtain the evolution of surface objects over time. These informations have important reference significance and are widely used to guide the detection and management of designated areas, such as environmental monitoring, disaster assessment, urban land planning, and map revision [2], [3], [4], [5].

Early work focused on the differential analysis of bitemporal images through professional knowledge and experience [1], such as visual interpretation and comparison methods. It has relatively reliable results but has expensive labor costs and is not suitable for large-scale high-resolution (HR) remote sensing images with complex spatial characteristics [6]. In order to save labor and time costs and realize CD automation, CD methods based on artificial intelligence have been developed rapidly. Machine learning-based methods, such as naive Bayesian [7], [8], random forest [9], [10], support vector machines [11], [12], and decision trees [13], are widely used in change detection of remote sensing images because of their excellent adaptability. But they still suffer from computational inefficiency. With the advancement and development of remote sensing technology, optical remote sensing images with very HR (VHR) are also widely used in many remote sensing computer vision tasks. These images have a higher spatial resolution, but they also present more difficulties and challenges for current CD techniques. First of all, for optical images showing large-scale regions, small-scale target changes are difficult to accurately detect due to the limitation of spatial resolution, which may lead to detection omissions. On the other hand, images at different times have different imaging conditions, including lighting conditions and atmospheric interference, which will cause brightness differences between images. This difference may lead to wrong semantic analysis and affect the final detection results. In addition, the detection targets in some images may change in shape and color over time, such as the demolition and reconstruction of houses, which will also increase the difficulty of remote sensing image CD. These problems require the existing CD model to have a more powerful performance of mining global high-level semantic information to enhance the sensing and spatial positioning of the detection target. At the same time, to effectively

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Yu Shangguan, Jinjiang Li, Yepeng Liu, and Fan Zhang are with the School of Computer Science and Technology, Shandong Technology and Business University, Yantai 264005, China (e-mail: liuyepengdream@gmail.com).

Caiming Zhang is with the School of Computer Science and Technology, Shandong University, Jinan 250100, China.

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obtain the fine information of large-scale targets, such as edges and shapes, to achieve high-precision distinction between targets and backgrounds, the ability to effectively combine and utilize multilevel features has also become important.

Convolutional neural networks (CNNs) in the field of deep learning have achieved automatic extraction of multilayer features and have demonstrated impressive performance in many computer vision tasks, including remote sensing image CD [14], [15], [16], [17], [18]. However, traditional CNN-based methods, limited by the size of the convolutional kernels, focus more on local perception of features, which makes it difficult to capture more effective global contextual information for images with long-range dependencies. Dilated convolution [19] and ResNet [20] can increase the receptive field of the model to enhance the global perception ability of the model, but they have a large number of parameters and high computational complexity. To obtain the features of interest directionally and increase the receptive field of the model, attention mechanisms, such as spatial attention [21], [22], [23], channel attention [21], [22], [23], [24], and self-attention [25], [26], [27], have become research hotspots. The use of attention enables the model to selectively focus on task-related features and capture the long-range dependencies between these features to better understand the semantic information of images. But simply using attention on a single feature to reweight it across channels or spatial dimensions is inefficient. The current transformer-based method shows a superior ability to explore global information, which uses multihead self-attention to model the long-range dependencies of different spatial information in images to enhance the understanding of global semantics and context. However, the traditional transformer-based method lacks the utilization of spatial information and fine-grained feature information and also has high computing requirements.

To effectively deal with the existing problems of HR remote sensing images, we design a change detection network based on transformer and attention. We believe that while utilizing features of multilevel granularity can help identify targets of different scales in images to prevent information loss, simply merging these features may overlook the potential connections between them. To strengthen the connection between these features and improve their fusion efficiency, we use attention fusion to focus and combine the content of interest to obtain representative features. In addition, we consider the respective characteristics of the transformer and CNN and propose a branch transformer block (BTB). BTB aims to strengthen the transformer's joint use of local spatial information and global semantics of the change feature to achieve accurate positioning and precise segmentation of changing targets.

Our specific contributions are summarized as follows.

1) We propose an attention filtering network based on a branch transformer (BTAF) to achieve accurate detection of changing content in bitemporal remote sensing images. Extensive experiments on typical change detection datasets verify its effectiveness against other state-of-the-art methods.
2) To effectively discover and retain the high-level semantics and low-level details of bitemporal features, we design a filtering structure based on attention fusion to screen out valuable information and retain it step by step. This structure is also applied to the fusion of multilevel difference features to enhance the feature representation of change information.
3) We propose a hybrid attention fusion module (HAFM) for the fusion of multiscale features, which simultaneously explores potential connections between different-level features in both spatial and channel dimensions to enrich the fused feature representation.
4) To strengthen the transformer's ability to learn and combine spatial information, a BTB is proposed. BTB has two input branches to incorporate semantic tokens and spatial features. BTB's branched multihead self-attention can effectively enhance self-attention learning and is suitable for large-size inputs.

In the rest of this article, the related work in Section II introduces the deep learning-based CD technology, attention mechanism, and transformer in detail. Section III presents the overall structure of our method and the implementation principles of different modules. Section IV describes the experimental part, including datasets, evaluation metrics, experimental results, and network visualization. The conclusion part is in Section V.

II. Related Work

A. Deep Learning-Based Change Detection Techniques

CD technology for HR remote sensing images that reflect rich geographic information and features has important significance for land monitoring and planning. Before deep learning methods were popularized, remote sensing CD mainly relied on traditional machine learning [28], [29], [30] and image processing [31], [32], [33] methods. They mainly acquire image features by manual design or complex algorithms, such as image algebra [31] and transformation [34]. With the advancement of remote sensing technology, these methods also exhibit significant limitations, such as being susceptible to seasonal changes, target shadows, and lighting conditions, resulting in reduced detection accuracy. Furthermore, the ability to automate manually designed features is limited due to the inherent reliance on expertise and experience. Current deep learning-based methods have the ability to automatically extract and process complex image features end-to-end and are widely used in advanced computer vision tasks, such as semantic segmentation [35], object detection [36], and image classification [37]. Compared with traditional CD methods, deep learning-based methods can more effectively learn high-level, discriminative features of raw data, which strengthens the understanding of complex scenes and high-level semantics in images. Deep learning-based CD methods can be roughly divided into three categories, including unsupervised methods, semisupervised methods, and supervised methods [38], which are divided according to the availability of labels for training data.

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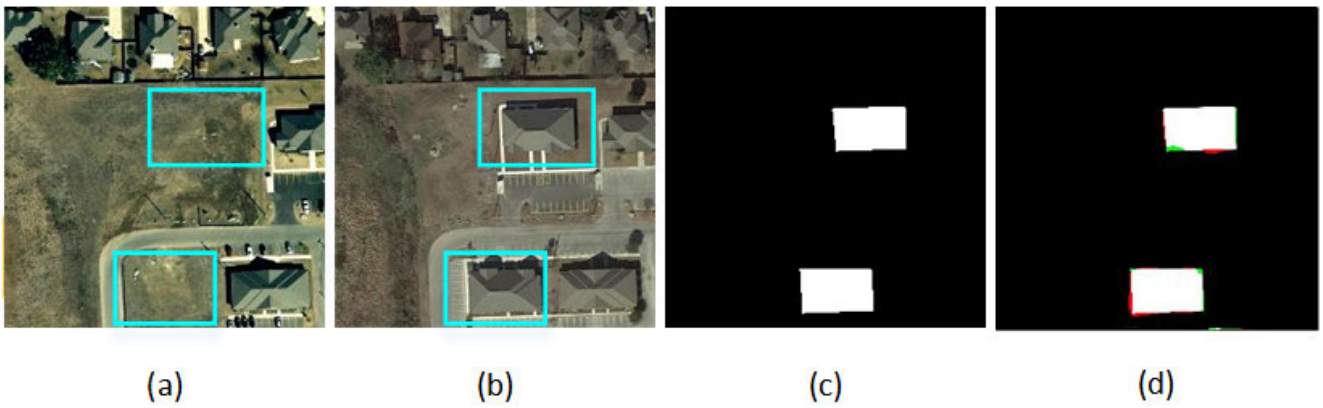


Fig. 1. Binary change detection mask obtained from bitemporal remote sensing images. (a) and (b) Images before and after the change, respectively. (c) Ground truth. (d) Detection result of our method.

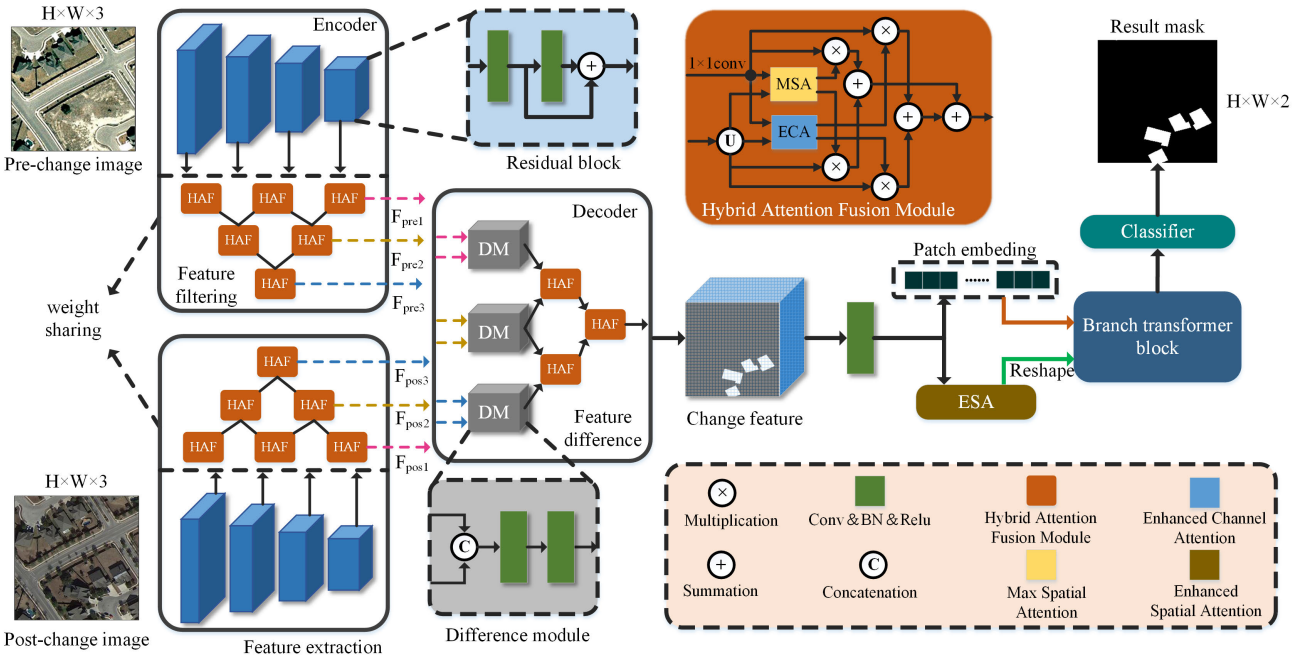


Fig. 2. Overview description of attention filtering network based on BTAF. Stacked residual blocks are first used to extract multiscale features. A HAFM is then used in Siamese's multilevel feature filtering and feature differentiation process to extract a fine change feature of bitemporal features. Then, we use the BTB to strengthen the exploration of the context and spatial characteristics of the change feature. Finally, a convolution operation is used as a classifier to obtain pixel-level detection results.

Unsupervised deep learning CD methods are mainly suitable for unlabeled remote sensing image datasets, which have more practical applications in the face of large-scale complex data than supervised learning using labels with expensive production costs. They automatically discover and learn the association between these data through the input training samples and provide valuable information for subsequent task processing [7]. Gong et al. [8] proposed a method combining sparse autoencoder, CNN, and unsupervised clustering algorithm to solve the problem of unsupervised triple change detection in synthetic aperture radar (SAR) images. This method uses a coefficient autoencoder to transform the log-ratio difference image to other feature spaces to capture key variations and suppress noise. The learned features are then clustered into three classes in an unsupervised manner as pseudolabels for a CNN-based variation feature classifier. Huang et al. [11] proposed a new unsupervised algorithm to detect building changes in HR multitemporal images. This algorithm first uses convolution operations to obtain low-level features of features, such as texture and spectral features and generates a combined low-level feature vector for each pixel. The deep belief network (DBN) is used in this algorithm to pretrain unlabeled samples. Afterward, its parameters are optimized by an extreme learning machine (ELM) classifier [12]. Yang et al. [9] proposed an unsupervised temporal distance–guided convolutional recurrent network. This network can utilize a time-distance-guided long short-term memory (TLSTM) unit to suppress the influence of spurious changes caused by shadows in remote sensing images.

The semisupervised approach is between supervised and unsupervised, which utilizes both labeled and unlabeled data to strengthen the training of the CD model. Semisupervised

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learning expands the training data through the use of unlabeled data to achieve better generalization capabilities of the model and can also help the model improve the performance of supervised learning. Wang et al. [10] used a graph-based knowledge supplementation network (GKSNet) to mitigate the noise impact of pseudolabels in self-supervised CD methods. Labeled datasets contain discriminative information that can be used as additional knowledge, and their extraction can effectively suppress noise. Based on this, the graph transfer module in GKSNet extracts contextual information from labeled data to target data to establish correlations between these datasets. Zhang and Shi [13] utilized a feature-difference CNN approach to achieve CD for HR RS images. This method uses CNN to learn deep features and transfer learning to obtain a dual-channel network that can generate multiscale feature difference maps, which share weights. A change magnitude guided loss function is used to help the network generate change magnitude maps while only using a small number of pixel-level samples.

Supervised deep learning uses labeled data to detect and calibrate training data, which effectively reduces the error between the final prediction result and the real label. Compared with the other two methods, the supervised method can obtain a more stable and accurate training model with a sufficient number of labels, which is very important for tasks requiring high precision [6]. Thanks to ResNet [20], deep convolution is possible and it can extract more complex and rich features, which makes the network structure based on deep CNN continuously improved to adapt to different tasks. Peng et al. [39] proposed an end-to-end UNet plus structure called UNet++ based on the structure of UNet [40] to effectively solve the accumulated errors of multistep CD. This method concatenates the paired RS images and fuses the global and local information of features in a densely connected way to obtain highly accurate CD results. Attention-based CD methods [21], [24], [41], [42], [43] can further optimize the model's context understanding ability for HR RS images and focus on valuable change information. Chen and Shi [25] proposed a Siamese spatial–temporal attention neural network to achieve high-precision detection on the publicly available HR dataset LEVIR-CD. They leverage a novel CD self-attention mechanism to model the spatiotemporal relationship of paired images, which can compute attention weights between pixels in different times and spaces to generate features with significant discriminative properties. Furthermore, the rational aggregation of multiscale features can effectively enhance the model's capture of global and local information. Li et al. [44] designed a lightweight, progressive feature aggregation, and supervised attention method for RS image CD. It fuses bitemporal multiscale feature information through a neighbor aggregation module and uses supervised attention to calibrate difference features to promote their progressive aggregation.

B. Transformer-Based CD Method

The transformer proposed by Vaswani et al. [45] was originally used in natural language processing (NLP) [46] tasks and has achieved excellent performance and has attracted much attention. Similarly, the improved vision transformer (Vit) [47] has played a huge role in many computer vision tasks, including CD [48], [49], [50], [51], [52], [53], [54], [55], [56], and has gradually become a hot trend. Vit abstractly divides images into semantic sequences and learns the interrelationships of these sequences through multihead self-attention to capture the contextual information between them to achieve better feature representation. Chen et al. [57] abstracted the interest changes in HR remote sensing images into some semantic tokens and modeled the contextual information of these tokens through a transformer encoder. A transformer decoder is used to restore the learned high-level tokens to fine features for high-quality detection. Zhang et al. [58] designed a U-shaped Swin transformer-based network to enhance the global exploration performance of the model. This Siamese network uses a multilevel Swing transformer block to build an encoder and decoder structure and achieve efficient CD. Wang et al. [59] proposed a method combining convolutional attention and transformer to solve the problem of difficult-to-recognize complex changes. It introduces a multiscale transformer in the traditional convolution to obtain more effective advanced features from bitemporal images. In [60], [61], [62], and [63], transformers are commonly used to obtain reasonable context and global dependencies, which compensate for the locality of CNN operations.

A large amount of complex scene information and shadows contained in HR RS images will interfere with the model's analysis and recognition of detection targets. However, over-attention to the high-level semantics and global context of features will also affect the accurate distinction of local details in the changing regions of interest in the image. In this article, we further explore the possibility of combining CNN with a transformer to jointly achieve an efficient fusion of local and global information.

III. Methodology

In this section, we first introduce the overall structure and loss function of our method. Our network structure mainly includes two modules, which are the HAFM and the BTB. They will be introduced in detail in the following. In addition, we also give the algorithm flow of our method as Algorithm 1.

A. Overall Framework of the Network

The shallow features obtained by CNN can preserve the local details of the image to a certain extent, while the deep features contain richer global information. However, the use of only a single level of deep features makes it difficult to balance the utilization of local details and global semantic information in related tasks of computer vision. Multilevel features contain feature information of different granularities that map different levels of abstract content, which can help reconstruct a complete and rich feature representation. How to reasonably combine the effective information of these homologous features becomes the key. Fig. 2 shows the overall architecture and implementation flow of our method. In this method, we first utilize residual blocks of different levels

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as the backbone of the network to extract multilevel features. To maximize the valuable information of these features, we design a funnel-like filtering structure of multilevel features based on the proposed HAFM. This encoder–decoder-like structure makes up for the limitation of each single-level feature by focusing on the meaningful information of multilevel features in a level-by-level manner. We obtained multiscale refined dual-temporal features through HAFM, and these features were introduced into the difference module (DM) for difference calculation and obtained difference features containing information of different granularities. We similarly use HAFM to filter out a representative change feature to reflect the changing information of features from different times. To further explore the contextual and spatial connections of changing features, we also propose a BTB to further optimize it. In the last part of the network, we use a convolutional classifier to obtain the final binary detection results.

Algorithm 1 Implementation Steps of BTAFNet Model

Input: $\mathbf{I_1}, \mathbf{I_2}$(*A pair of satellite remote sensing images at different times*)
Output: O
// *step*1: *Obtain multi-scale features through stacked convolutional residual blocks.*
for i *in* $\{1, 2\}$ **do**
  $\mathbf{F_{i,1}} = ConvBlock(\mathbf{I_i})$
  for j *in* $\{2, 3, 4\}$ **do**
    $\mathbf{F_{i,j}} = ConvBlock(\mathbf{F_{i,j-1}})$
  end
// *step*2:*Feature filtering using multi-level HAFM*
  for m *in* $\{1, 2, 3, \}$ **do**
    $\mathbf{F'_{i,m}} = HAFM(\mathbf{F_{i,m}}, \mathbf{F_{i,m+1}})$
  end
  for n *in* $\{1, 2\}$ **do**
    $\mathbf{F''_{i,n}} = HAFM(\mathbf{F'_{i,m}}, \mathbf{F'_{i,m+1}})$
  end
  $\mathbf{F'''_i} = HAFM(\mathbf{F''_{i,1}}, \mathbf{F''_{i,2}})$
end
// step3 : *Fusion of multi-level bitemporal features to obtain a fine change feature*
$\mathbf{F_{c,1}} = \mathrm{DM}(\mathbf{F'_{1,3}}, \mathbf{F'_{2,3}})$
$\mathbf{F_{c,2}} = \mathrm{DM}(\mathbf{F''_{1,2}}, \mathbf{F''_{2,2}})$
$\mathbf{F_{c,3}} = \mathrm{DM}(\mathbf{F'''_1}, \mathbf{F'''_2})$
for k *in* $\{1, 2\}$ **do**
  $\mathbf{F'_{c,k}} = HAFM(\mathbf{F_{c,k}}, \mathbf{F_{c,k+1}})$
end
//*step*4 : *Use BTB to optimize change feature*
$\mathbf{F_b} = BTB(ESA(\mathbf{F_c}), Embeding(\mathbf{F_c}))$
//*step*4 : *Use convolution to obtain binary results*
$\mathbf{O} = Classifier(\mathbf{F_c})$

B. Hybrid Attention Fusion Module

For the same input image, there is a gradually transforming dependency between its features at different levels. To explore their interdependence and correlation and achieve an efficient fusion of features, we propose an HAFM, as shown in Fig. 3. HAFM combines enhanced channel attention (ECA) and maximum spatial attention to aggregate feature information from different dimensions and captures their interrelationships dynamically. This correlation can help to accurately locate key feature information and adaptively adjust the weights of different parts. HAFM guides the information redistribution and fusion of input features through the learned weights to obtain effective feature representations. Compared with using existing channel attention or spatial attention for a single feature to enhance feature representation, HAFM combines the advantages of both and uses them to explore the interspatial and interchannel relationships between input features. This will facilitate the effective fusion of multilevel features and enrich the fused feature representation. Specifically, we first process the two input features to ensure that they have the same size and number of channels. Given the input $F_l \in \mathbb{R}^{C_l \times H_l \times W_l}$ and $F_h \in \mathbb{R}^{C_h \times H_h \times W_h}$, where C_l and C_h represent the number of channels of low-level feature F_l and high-level feature F_h, respectively, and H_l, W_l, H_h, and W_h represent the height and width of the two features respectively. We use the formula to express this preprocessing process as

$$\begin{aligned} F'_h &= \mathrm{Up}(F_h) \\ F'_l &= \mathrm{Conv}_1(F_l) \end{aligned} \tag{1}$$

where $\mathrm{Up}(\cdot)$ represents the upsampling operation, $\mathrm{Conv}_1(\cdot)$ represents the 1×1 convolution layer. The dimensions of F'_h and F'_l are both $\mathbb{R}^{C_h \times H_l \times W_l}$. Afterward, we send two feature maps of the same shape to two attention branches, namely, ECA and max spatial attention (MSA). Through these two attention branches, we learn the interfeature dependencies in the spatial and channel dimensions, respectively, and derive a weight matrix that reflects the importance of each pixel and channel in the input features. Specifically, in the ECA branch, we first use AvgPool and MaxPool operations, respectively, to compress the spatial information of the two features in order to focus only on the channel information. They are then concatenated to obtain a feature with a dimension of $\mathbb{R}^{4C_h \times 1 \times 1}$. We use a 1×1 convolutional layer to reduce the number of channels from $4C_h$ to C_h and utilize consecutive fully connected layers to enhance nonlinear expression capabilities. The sigmoid operation is used to obtain the channel attention weight matrix α. After that, we perform element-level multiplication of F'_l and F'_h with α and $1-\alpha$, respectively, and fuse them by an addition operation

$$\begin{aligned} F_l^{ch} &= \mathrm{Concat}\big(\mathrm{AvgPool}\big(F'_l\big), \mathrm{MaxPool}\big(F'_l\big)\big) \\ F_h^{ch} &= \mathrm{Concat}\big(\mathrm{AvgPool}\big(F'_h\big), \mathrm{MaxPool}\big(F'_h\big)\big) \\ F_c^{ch} &= \mathrm{Conv}_1\big(\varphi\big(\omega\big(\mathrm{Concat}\big(F_l^{ch}, F_h^{ch}\big)\big)\big)\big) \\ \alpha &= \mathrm{Sigmoid}(F_c) \\ F_{ch} &= F'_l \cdot \alpha + F'_h \cdot (1-\alpha) \end{aligned} \tag{2}$$

where $\omega(\cdot)$ and $\varphi(\cdot)$ represent two fully connected layers, α represents a weight vector with a dimension of $\mathbb{R}^{C_h \times 1 \times 1}$, and $F_{ch} \in \mathbb{R}^{C_h \times H_l \times W_l}$ represents the output feature of the ECA branch. On the other hand, in the MSA branch, we perform max operations on F'_l and F'_h, respectively, along the channel

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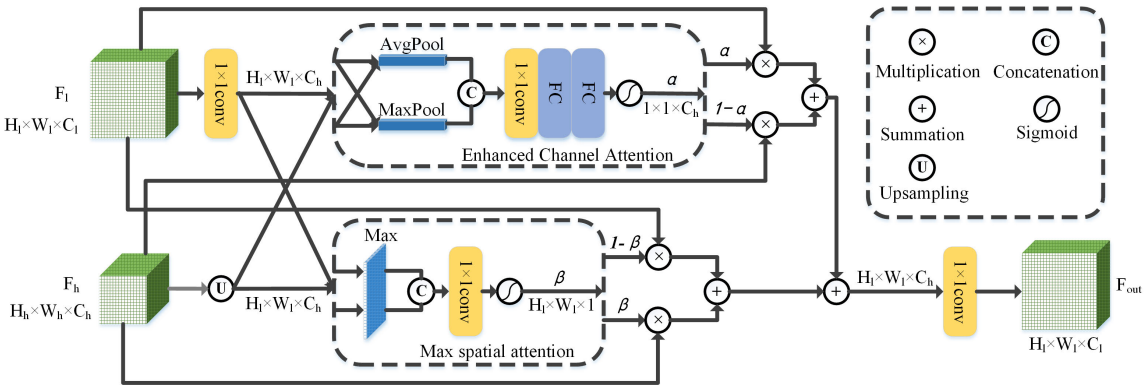


Fig. 3. Description of the HAFM, including ECA and maximal spatial attention to fuse channel and spatial information of features at different scales.

axis to compress their channel information and obtain two intermediate features with a dimension of $\mathbb{R}^{1\times H_l\times W_l}$. Then, we concatenate these features and use a 1×1 convolution operation to adjust the number of channels to 1. Similarly, the sigmoid operation is used to calculate the importance probability of each spatial location and obtain a spatial weight matrix β. This matrix reweights the F_l' and F_h' according to the attention of different spatial regions. This process is expressed as

$$\begin{aligned} F_c^{sp} &= \mathrm{Conv}_1\left(\mathrm{Concat}\left(\mathrm{Max}\left(F_l'\right), \mathrm{Max}\left(F_h'\right)\right)\right) \\ \beta &= \mathrm{Sigmoid}\left(F_c^{sp}\right) \\ F_{sp} &= F_l' \cdot (1-\beta) + F_h' \cdot \beta \end{aligned} \tag{3}$$

where β represents a spatial weight matrix with dimension $R^{1\times H_l\times W_l}$ and $F_{sp} \in \mathbb{R}^{C_h\times H_l\times W_l}$ represents the output feature of the MSA branch. After obtaining the results of the two attention branches, we fuse them using elementwise addition, and finally, use a 1×1 convolution layer to adjust the number of channels from C_h to C_l to reduce the number of parameters. This process is described by the following formula:

$$F_{\text{out}} = \mathrm{Conv}_1\left(F_{sp} + F_{ch}\right). \tag{4}$$

C. Branch Transformer Block

After acquiring the change feature, it is necessary to enhance the performance of context mining at this stage in order to further capture and locate change targets. The vision transformer is known for its powerful context aggregation and global awareness in various computer vision tasks. However, its computational requirements are closely related to the input feature size, which makes it less suitable for remote sensing images with HR. In addition, the single use of multihead self-attention to capture long-range connections in the change feature is likely to cause the model to ignore spatial details and affect the accurate segmentation of changing objects. In order to solve these problems, we propose a BTB that combines the advantages of transformer and CNN, and its structure is shown in Fig. 4. The input of BTB contains two branches, one is the tokens obtained by the embedding layer, and the other is the refined spatial feature obtained by the ESA (ESA) module. BTB adopts a parallel branching strategy and further processes these inputs by branch multihead self-attention (BMSA) to realize their effective fusion. This will help the model to effectively combine the spatial details of features while strengthening the global context modeling ability. In addition, in order to reduce the computational burden of the model and increase the receptive field of self-attention, the BMSA in BTB reduces the input token sequence into two pairs of key and value of different sizes through sequence reduction [64] to match the query. Specifically, given an input change feature $F \in R^{H\times W\times C}$, where C denotes the number of channels, H and W denote the width and height of the feature, respectively. We first use the embedding layer to divide F into tokens $T \in R^{HW\times C}$ and obtain the spatial feature vector (SFV) $V_s \in R^{HW\times C}$ through an ESA module and reshape operation. T and V_s are then fed into a BMSA, as shown in Fig. 4(b). In BMSA, we use a single-linear-layer to obtain the query and employ a combination of stride convolution and linear-layer operations to acquire key maps and value maps of different sizes, enabling hierarchical mapping. This process can be expressed as follows:

$$\begin{aligned} Q &= TW_q \\ K_i &= \varepsilon_i(T)W_k \\ V_i &= \zeta_i(T)W_v \end{aligned} \tag{5}$$

where K_i and V_i $(i = 1, 2)$ represent the obtained key maps and value maps of different sizes, respectively. W_q, W_k, and W_v denote the learnable parameters of the linear projection layers, respectively. $\varepsilon_i(\cdot)$ and $\zeta_i(\cdot)$ $(i = 1, 2)$ denote convolution operations with different strides. The implementation process of a branch self-attention head can be expressed as

$$Y_i = \mathrm{SA}(Q, K_i, V_i) = \sigma\left(\frac{QK_i^T}{\sqrt{d_{\text{head}}}}\right)V_i \tag{6}$$

where $\sigma(\cdot)$ represents the softmax operation. Afterward, we concatenate the SFV V_s with different self-attention branches Y_i $(i = 1, 2)$ separately and use linear layers to blend contextual content and valuable spatial information. BMSA combines the results of two branches through elementwise addition to further enhance self-attention learning. The following formula represents this process:

$$\begin{aligned} M_i &= \mathrm{Linear}(\mathrm{concate}(Y_i, V_s)) \\ N &= \mathrm{Linear}(M_1 + M_2) \end{aligned} \tag{7}$$

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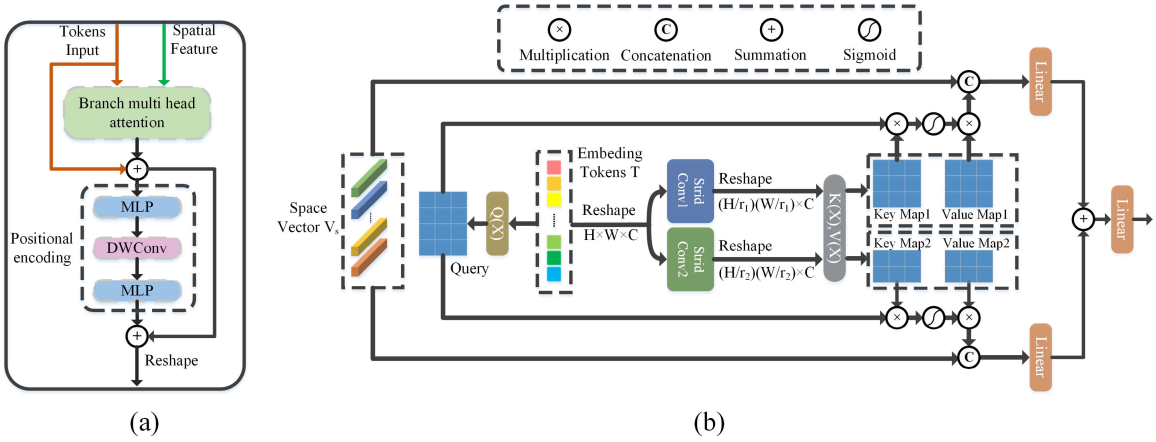


Fig. 4. (a) Description of the BTB. (b) Description of the BMSA. The input consists of semantic tokens obtained by the embedding layer and spatial vectors obtained by ESA.

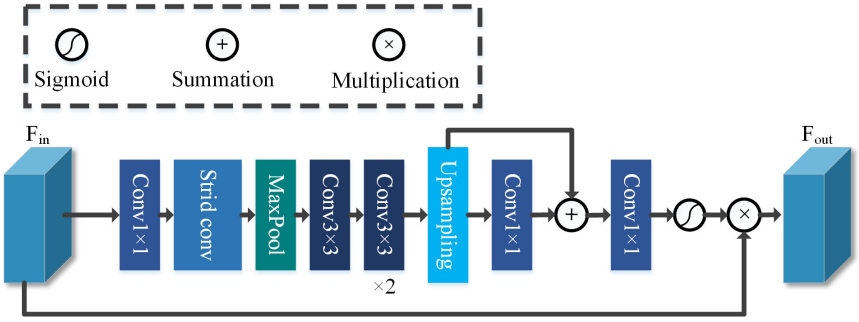


Fig. 5. Description of the ESA module.

where M_i $(i = 1, 2)$ represents the mixed vector that combines global context and local spatial information and N represents the output of BMSA. After this, we follow the positional encoding method in ChangeFormer [65]. It employs two MLP layers and a 3×3 depthwise convolutional layer for positional encoding, which is expressed by the formula

$$Z = \mu(G(DW(\mu(N + T)))) + N \tag{8}$$

where $\mu(\cdot)$ represents the MLP layer, $DW(\cdot)$ represents the 3×3 depthwise convolution operation, and G represents the Gaussian Error Linear Unit activation. Finally, we reshape the output obtained from BTB to the shape of $R^{C\times H\times W}$ and obtain the final binary detection result Z through a convolution-based classifier.

D. Other Network Details

1) Loss Function: We optimize the model during the training phase by minimizing the cross-entropy loss, which is formally defined as follows:

$$L = \frac{1}{H_0 \times W_0} \sum_{h=1,w=1}^{H,W} l(P_{hw}, Y_{hw}) \tag{9}$$

where H_0 and W_0 represent the height and width of the original image, respectively, $l(P_{hw}, Y) = -\log(P_{hwy})$ is the cross-entropy loss, and Y_{hw} is the label of the pixel at (h, w) position.

2) Enhanced Spatial Attention Module: For HR remote sensing images, in order to achieve more subtle change segmentation, such as the edge and shape of the changing target, it is necessary to preserve the valuable spatial information of the change feature. To this end, we extend a branch that introduces an ESA module to strengthen the spatial awareness of the change feature and focus on important spatial content before feeding the change feature into BTB. The principle of ESA is shown in Fig. 5. Specifically, for the input $F_{\text{in}} \in R^{C\times H\times W}$, a 1×1 convolution is first used to reduce the number of channels of F_{in} from C to $C/4$ to compress the channel information. A 3×3 stride convolutional layer and max pool layer are used to extract local spatial features of the change feature and achieve feature downsampling. To obtain richer long-range spatial connections, a set of 3×3 convolutional sequences with a stride of 1 is used to further explore deep features. ESA then upsamples the resulting feature map to its original size and uses a 1×1 convolutional layer to restore its number of channels to C. Finally, the sigmoid function is used to obtain the spatial attention weights, which are elementwise multiplied with the input to obtain the spatial attention features $F_{\text{out}} \in R^{C\times H\times W}$.

3) Difference Module: We introduce three simple DMs in our method to obtain multiscale difference features between the bitemporal fine features obtained from the Siamese backbone network. These difference features are then filtered by HAFM to obtain an effective change feature. Specifically,

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TABLE I
QUANTITATIVE COMPARISON RESULTS (%) BETWEEN OUR METHOD AND OTHER STATE-OF-THE-ART METHODS ON LEVIR-CD, THE BEST EVALUATION SCORES HAVE BEEN MARKED IN BOLD

Method	LEVIR-CD				
	Precision	Recall	F1	IoU	OA
FC-EF	86.91	80.17	83.40	71.53	98.39
DTCDSCN	88.53	86.83	87.67	78.05	98.77
IFNet	**94.02**	82.93	88.13	78.77	98.87
SNUNet	89.18	87.17	88.16	78.83	98.82
BIT	89.24	89.37	89.31	80.68	98.92
ChangeFormer	92.05	88.80	90.40	82.48	99.04
MSCANet	89.79	87.57	88.67	79.64	98.86
PaFormer	90.75	87.23	88.95	88.95	98.86
ICIF	92.07	89.30	90.81	83.31	99.08
BTAFNet(Ours)	92.81	**90.52**	**91.65**	**84.59**	**99.16**

TABLE II
QUANTITATIVE COMPARISON RESULTS (%) BETWEEN OUR METHOD AND OTHER STATE-OF-THE-ART METHODS ON WHU-CD, THE BEST EVALUATION SCORES HAVE BEEN MARKED IN BOLD

Method	WHU-CD				
	Precision	Recall	F1	IoU	OA
FC-EF	71.63	67.25	69.37	53.11	97.61
DTCDSCN	63.92	82.30	71.95	56.19	97.42
IFNet	**96.91**	73.19	83.40	71.52	98.83
SNUNet	85.60	81.49	83.50	71.67	98.71
BIT	86.64	81.48	83.98	72.39	98.75
ChangeFormer	90.50	79.61	84.51	73.18	98.59
MSCANet	90.81	83.79	87.16	77.24	98.81
PaFormer	85.99	**93.77**	89.71	81.34	98.96
ICIF	92.93	88.70	90.77	83.09	99.13
BTAFNet(Ours)	94.33	91.88	**93.09**	**87.07**	**99.34**

DM first concatenates two input features in the channel dimension through the concatenate operation and obtains a merged feature with twice the original number of channels. Then, DM uses successive 3×3 convolution operations to learn feature variation and recover the number of channels. Finally, DM uses the ReLU activation function and BatchNormal processing to obtain the difference features. This process can be summarized as

$$F_{\text{diff}} = B\left(R\left(\text{Conv}_{3\times3}\left(\text{Concat}\left(F_{\text{pre}}, F_{\text{pos}}\right)\right)\right)\right) \tag{10}$$

where F_{pre} represents the feature before the change, F_{pos} represents the feature after the change, R represents the ReLU activation function, and B represents BatchNormal.

IV. EXPERIMENTS

In this part, we will introduce the remote sensing image CD dataset used, the evaluation metrics, the comparison experiment between the experiment and the state-of-the-art method, and the ablation experiment of different modules.

A. Datasets

In this article, we use three public HR datasets for comparison and ablation experiments, including LEVIR-CD [25], WHU-CD [66], and GZ-CD [67]. With these datasets, we present qualitative and quantitative experimental results to illustrate the effectiveness of our method.

TABLE III
QUANTITATIVE COMPARISON RESULTS (%) BETWEEN OUR METHOD AND OTHER STATE-OF-THE-ART METHODS ON GZ-CD, THE BEST EVALUATION SCORES HAVE BEEN MARKED IN BOLD

Method	GZ-CD				
	Precision	Recall	F1	IoU	OA
FC-EF	77.49	65.74	71.13	55.19	94.89
DTCDSCN	**88.19**	78.38	83.00	70.93	96.92
IFNet	55.70	67.41	67.00	43.99	91.74
SNUNet	87.55	80.05	83.64	71.84	97.00
BIT	82.40	78.18	80.23	66.99	96.31
ChangeFormer	84.59	65.23	73.66	58.30	95.53
MSCANet	86.65	76.43	81.22	68.38	96.61
PaFormer	86.25	77.95	81.89	69.34	96.70
ICIF	87.57	82.31	84.86	73.70	97.19
BTAFNet(ours)	87.86	**83.73**	**85.75**	**75.05**	**97.33**

LEVIR-CD [25] is a large-scale public dataset obtained from Google Earth. It contains 637 pairs (0.5 m/pixel) of dual-time-phase remote sensing images at 1024×1024 resolution. The dataset contains images spanning from 5 to 14 years that mainly reflect land use changes, such as new construction and demolition of buildings, including apartment houses, small garages, and large warehouses, in urban and suburban areas. These diachronic images are annotated using binary labels (0 for no change and 1 for change). In order to visualize the detection results more clearly and to enhance the detection efficiency, we randomly cropped the 637 pairs of data into nonoverlapping images of 256×256 resolution and delineated them in the ratio of 2:7:1 for testing/training/validation.

WHU-CD [66] is a dataset documenting urban reconstruction since the February 2011 earthquake of magnitude 6.3 occurred. An aerial image with a resolution of $32\,507 \times 15\,354$ (0.075 m/pixel) contains 16 077 buildings in 20.5 square kilometers. We similarly obtained 6096/762/762 pairs of subimages with a resolution of 256×256 for training/validation/testing by random cropping and image enhancement.

GZ-CD [67] contains HR satellite images of remotely sensed CDs of the Guangzhou area of China for all seasons over a 13-year period from 2006 with a spatial resolution of 0.55/pixel. It recorded a total of 19 pairs of aligned images with image resolutions ranging from 1006×1168 pixels to 4936×5224. The same as the other two datasets, we cropped them into 256×256 nonoverlapping images by random cropping and trained/tested/validated them according to the ratio of 8:1:1.

B. Implementation Details and Evaluation Metrics

1) Implementation Details: We built our overall model on PyTorch version 1.8 and trained it with a single NVIDIA RTX TITAN GPU with 24-GB memory. We use AdamW with weight decay correction as an optimizer and set the weight decay to 0.01 and beta to (0.9, 0.999). The training batch size

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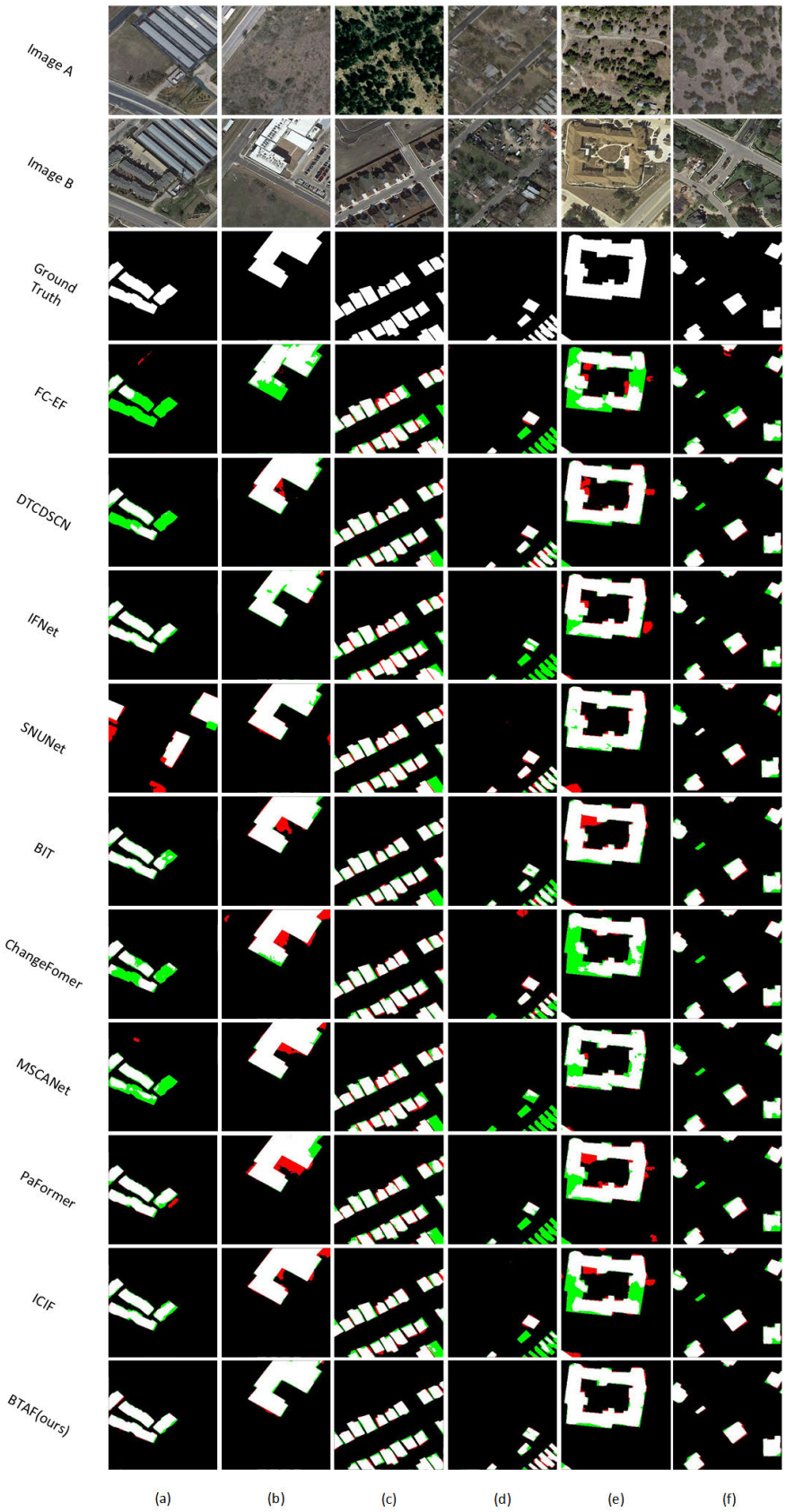


Fig. 6. Visual comparison results of BTAF and other advanced methods on LEVIR-CD. Different colors in the figure express different detection contents, among which white indicates true positive, black indicates true negative, red indicates false positive, and green indicates false negative. (a)–(f) Visual comparison results on different images selected from the test set of LEVIR-CD.

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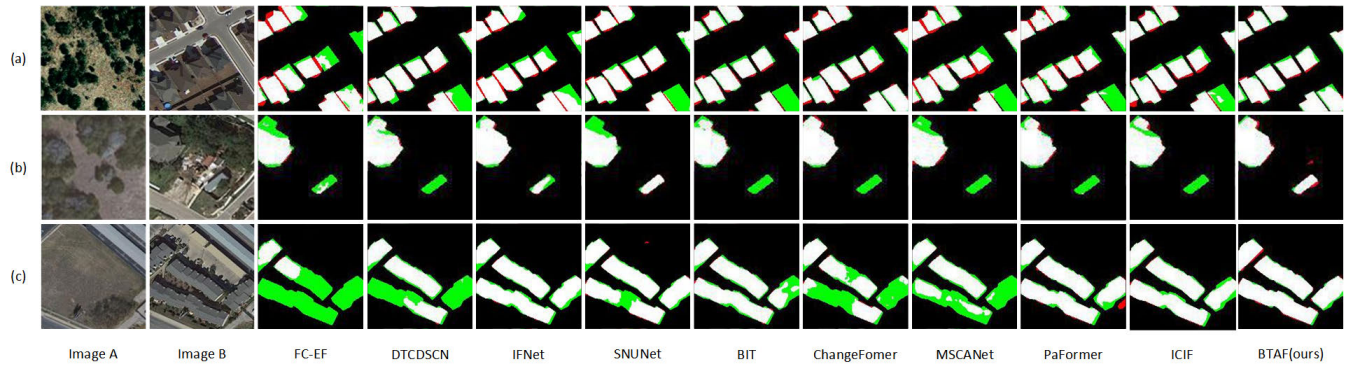


Fig. 7. More details of the visual comparison results of our method and other state-of-the-art methods on LEVIR-CD. (a) Corresponds to Fig. 6(c). (b) Corresponds to Fig. 6(f). (c) Corresponds to Fig. 6(a).

is 4 and the initial learning rate is 0.0001 which is decayed to 0 by linear decay at 200 epochs of training. In addition to this, we augmented the training images with data by means of Gaussian blurring, random cropping, random flipping, random color dithering, and random rescaling in order to strengthen the model's generalization ability.

2) Evaluation Metrics: In comparison with other state-of-the-art methods for quantitative experiments, we chose to use pixel-based F1-score and intersection over union (IoU) of the change category as the main evaluation metrics. In addition, we also use precision, recall, and overall accuracy (OA) as additional evaluation metrics. These metrics are defined as follows

$$\begin{aligned} \mathrm{F1} &= 2\frac{\mathrm{Precision}\cdot\mathrm{Recall}}{\mathrm{Precision}+\mathrm{Recall}} \\ \mathrm{Precision} &= \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \\ \mathrm{Recall} &= \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \\ \mathrm{IoU} &= \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}+\mathrm{FP}} \\ \mathrm{OA} &= \frac{\mathrm{TP}+\mathrm{PN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FN}+\mathrm{FP}} \end{aligned} \tag{11}$$

where TP denotes the number of true positive, TN denotes the number of true negative, FN denotes the number of false negative, and FP denotes the number of false positive.

C. Comparison With State-of-the-Art Methods

1) State-of-the-Art Methods: Nine state-of-the-art methods are used for quantitative and qualitative comparison experiments with our method. These methods include pure CNN methods [14], attention-based methods [22], [23], [68], transform-based methods [57], [61], [65], and methods based on transformer combined with CNN [69], [70]. Their principles are briefly described as follows.

1) *FC-EF [14]:* A CD network with FCN and UNet as the basic structure. This method uses a single branch of convolution to detect changes after stitching bitemporal images.
2) *DTCDSCN [22]:* A deep convolutional approach incorporating multitask constraints to enhance discrimination and segmentation of varying objects. It additionally uses dual attention to optimize feature representation.
3) *IFNet [23]:* A deep supervised image fusion method based on attention and FCN, which obtains paired depth features through FCN and uses attention to fuse the depth features and difference features of bitemporal images.
4) *SNUNet [68]:* A method that uses a Siamese dense connection structure to enhance the acquisition of feature localization information. It also proposes a channel of attention for deep supervision.
5) *Bitemporal Image Transformer (BIT) [57]:* A transformer-based method that treats interesting changes in the image as semantic tags and implements context modeling of the semantic tags through a transformer encoder–decoder.
6) *Changeformer [65]:* A Siamese-structured transformer-based approach. It uses a transformer encoder and multilayer perception (MLP) decoder to achieve high-precision CD.
7) *MSCANet [69]:* A method based on CNN and transformer applied to farmland CD. It uses CNN to extract effective features and uses a transformer to aggregate context information for features.
8) *PaFormer [61]:* An end-to-end approach based on prior-aware transformers. This method uses an a priori translator to obtain prior structural information about changing buildings and an a priori perceptual transformer to extract contextual labels based on prior features.
9) *ICIF [70]:* A transformer and CNN-based approach that effectively exploits the combined potential of transformer and CNN in feature extraction.

2) Results of Comparative Experiments: In this section, state-of-the-art methods are compared with our method to qualitatively and quantitatively demonstrate the effectiveness of our method. Tables I–III show the results of quantitative comparisons between our method and other methods on the LEVIR-CD, WHU-CD, and GZ-CD, datasets, respectively. Figs. 6–11 show the visual comparison results and details of these methods and our method. For the rationality of comparison, we choose some typical test images as the objects of comparison.

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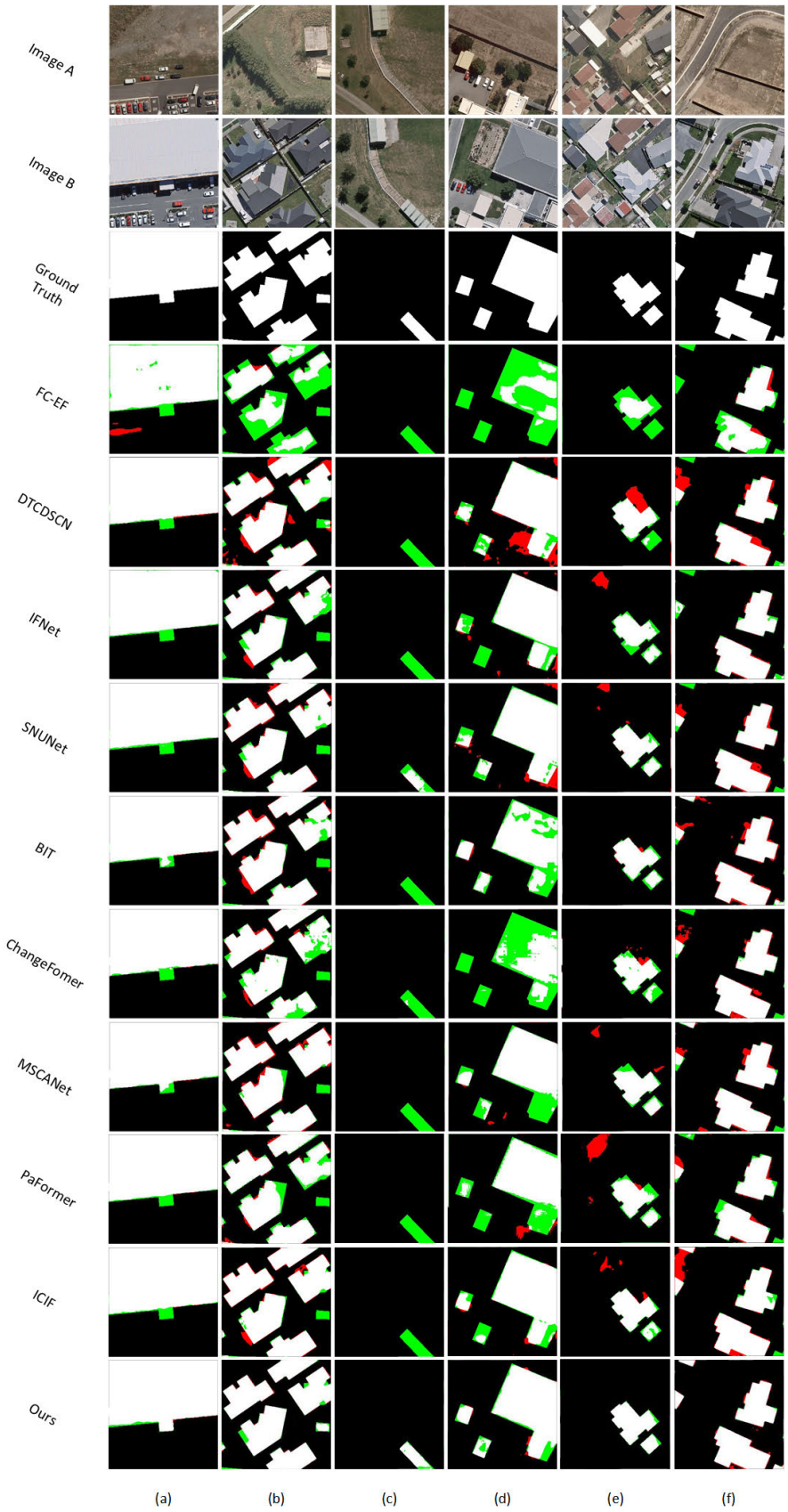


Fig. 8. Visual comparison results of BTAF and other advanced methods on WHU-CD. Different colors in the figure express different detection contents, among which white indicates true positive, black indicates true negative, red indicates false positive, and green indicates false negative. (a)–(f) Visual comparison results on different images selected from the test set of WHU-CD.

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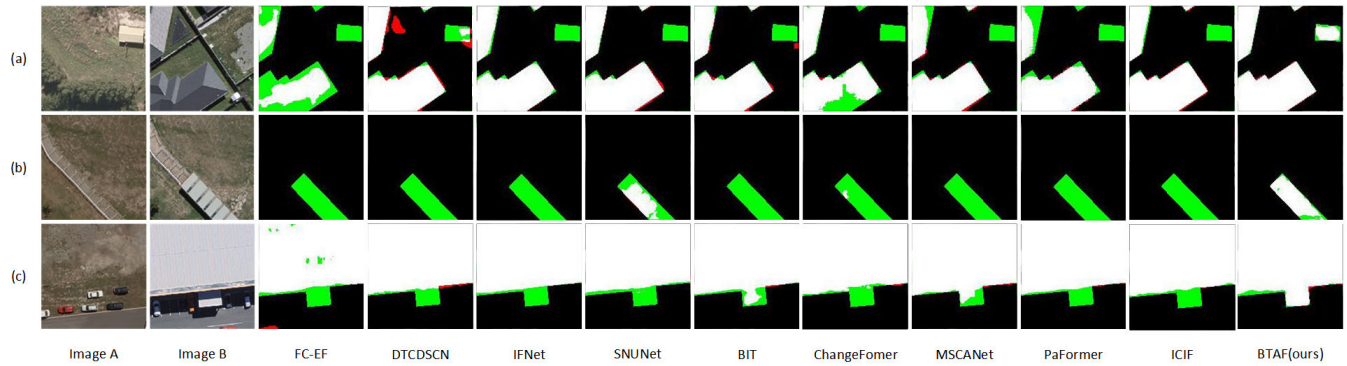


Fig. 9. More details of the visualization comparison results between our method and other state-of-the-art methods on WHU-CD. (a) Corresponds to Fig. 8(b). (b) Corresponds to Fig. 8(c). (c) Corresponds to Fig. 8(a).

As shown in Table I, our method achieves the best performance on LEVIR-CD except for precision. Specifically, our method achieves 91.65%/84.59%/99.16% F1-score/IoU/OA, which is 0.84%/1.28%/0.08% higher than the suboptimal ICIF based on transformer and CNN, respectively. The index of recall achieved 90.52%, 1.15% higher than the second-ranked BIT. As can be seen from Fig. 6, our method effectively reduces the false positive (red) and false negative (green) regions in the HR LEVIR-CD dataset, thus showing better CD accuracy. From Fig. 6(a)–(f), our method significantly outperforms other state-of-the-art methods in the overall detection performance of images. In Fig. 6(c), (d), and (f), it is noticed that most of the methods except our method suffer from the loss of detected objects (shown in green). In order to easily observe this detection difference, we selected representative images and enlarged their details, as shown in Fig. 7. Specifically, it can be seen from Fig. 7(a) that a building object in the lower right corner is difficult to detect by most methods because it has a similar color to the foundation and an inconspicuous shape. Our method can accurately localize it and achieve higher precision discrimination. The tiny target in the middle part of Fig. 7(b) is affected by the cluttered environment and background factors and is easily overlooked by most methods. Our method shows a good detection effect. Fig. 7(c) shows that our method has better detail handling for complex objects.

Table II shows the quantitative comparison on WHU-CD. Our method achieves the highest 93.09%/87.07%/99.34% on F1-score/IoU/OA indicators, which are 2.32%/3.98%/0.21% higher than ICIF, respectively. IFNet and PaFormer achieve the best results on precision and recall, respectively. Fig. 8 depicts the comparison results visualized on WHU-CD and Fig. 9 shows more details, our method significantly achieves the most superior detection performance. From 8(a)–(f), our method recognizes all detected objects and their shapes as completely and accurately as possible compared with other methods. Taking Fig. 9(a) as an example, the small buildings on the upper right of the picture have the same color as the ground and are difficult to distinguish, which leads to the detection loss of other methods except our method. Fig. 9(b) also shows a similar problem. The detection target in the visualization result picture is consistent with the road trajectory, so most of the methods miss the detection. The entrance part of the large building in Fig. 9(c) is affected by the shadow and the large building body appears to be separated, which makes it look more like a noninteresting changing vehicle next to it. Our method smoothly solves this problem and achieves correct understanding and recognition of detection objects.

The quantitative results in Table III show the comparison between our method and other methods in GZ-CD. Our method achieves the best results on all metrics except Precision. Specifically, our method is 0.89%/1.42%/1.35%/0.14% higher than the second-ranked ICIF in F1-score/Recall/IoU/OA, respectively. Figs. 10 and 11, respectively, show the visualization results of qualitative comparison experiments on the GZ-CD dataset and more details in it. Our method still achieves the best overall performance. According to Fig. 10(b), (c), and (f), our method performs relatively better in dealing with the edges and structures of the content of interest and effectively avoids the structure loss of changing objects. In addition, in Fig. 11(c), most of the methods wrongly determine the top-left region as the changed object, and some methods are accompanied by more false positive regions. This may be caused by subtle differences in brightness and color conditions between the changed and nonchanged regions of interest. Our method successfully avoids this misjudgment and exhibits a better understanding of the background and environment of the detected object.

The results in Tables I–III and Figs. 6–11 show that our method can achieve outstanding performance in the HR remote sensing image CD task and has advantages over state-of-the-art methods. Specifically, our method has a more effective ability to deal with complex changes, which can reduce the interference from the scene and environment around the detection target to achieve a more complete detection effect. On the other hand, our method exhibits better details when dealing with irregularly changing objects. This may be because our method can more effectively combine the global and local information of changing features by branching the transformer block to accurately locate the detection target and improve its structure. Attention filtering for multilevel bitemporal features also helps to strengthen the feature representation of interesting changes to slow down the interference of unimportant changes. Table IV presents the number of model parameters (Params)

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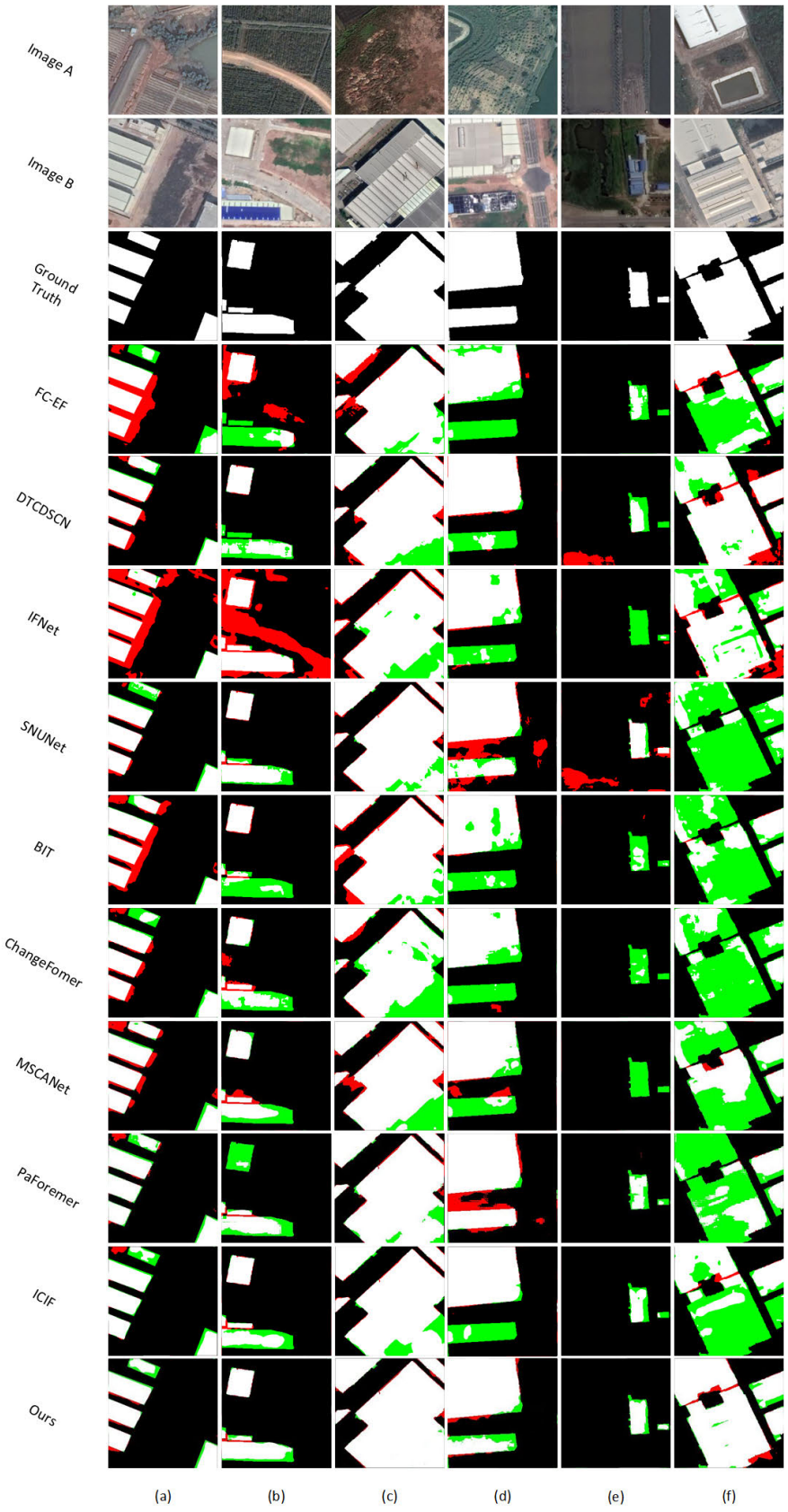


Fig. 10. Visual comparison results of BTAF and other advanced methods on GZ-CD. Different colors in the figure express different detection contents, among which white indicates true positive, black indicates true negative, red indicates false positive, and green indicates false negative. (a)–(f) Visual comparison results on different images selected from the test set of GZ-CD.

and floating point operations (FLOPs) of different models to illustrate their computational efficiency. It can be seen from Table IV that our method has fewer Params and is only higher than FC-EF and BIT but has higher FLOPs and is only lower

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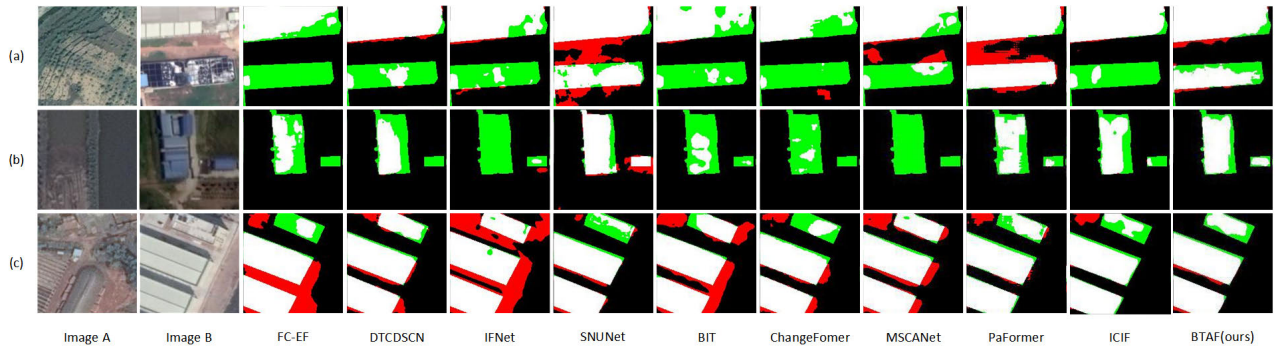


Fig. 11. More details of the visualization comparison results of our method and other state-of-the-art methods on GZ-CD. (a) Corresponds to Fig. 10(d). (b) Corresponds to Fig. 10(e). (c) Corresponds to Fig. 10(a).

TABLE IV
COMPARISON OF FLOPS AND PARAMS OF OUR PROPOSED METHOD WITH OTHER STATE-OF-THE-ART METHODS

Method	FLOPs(G)	Params(M)
FC-EF	3.57	1.35
FC-Siam-Di	4.72	1.35
FC-Siam-Conc	5.32	1.55
DTCDSCN	13.21	31.26
IFNet	82.26	50.44
SNUNet	54.82	12.03
BIT	10.59	3.49
ChangeFormer	21.18	29.73
MSCANet	14.77	16.42
PaFormer	10.86	16.13
ICIF	25.36	23.82
BTAFNet(ours)	61.72	9.48

than IFNet. In order to obtain more accurate detailed features, BTAF focuses on extracting low-level multigranularity features and further obtains high-level semantic information by branching the transformer block, which also results in a higher cost of computing resources. In general, although our method has fewer Params, it still needs to explore more lightweight structures to speed up model training.

D. Ablation Experiments

In order to verify the effectiveness of the proposed HAFM and BTB in our method, we performed ablation experiments on them on LEVIR-CD, WHU-CD, and GZ-CD, respectively, and reported the results of F1-score and IoU as Table V and their visualization results are shown in Fig. 12. Among them, baseline means that the baseline model of HAFM and BTB is eliminated, and it uses element-level addition to replace the role of HAFM in the feature extraction and filtering stages.

1) Ablation on HAFM: HAFM is used in the early feature extraction stage of the model to cooperate with the residual block to strengthen the exploration of potential connections between multiscale features and effectively combine and retain them. Our model mainly implements the extraction and filtering of efficient features for bitemporal images through the hierarchical structure of HAFM. No. 2 in Table V shows the effect of adding HAFM alone on the baseline. Specifically, after adding HAFM, the model obtained 90.24%/90.11%/82.16% F1-score on the three datasets, which were 2.15%/6.24%/3.99% higher than the baseline. As for IoU, they are 3.04%/10.36%/4.67% higher, respectively. On the other hand, according to the results in Fig. 12, we can see that when HAFM is added to the Baseline alone, the model can achieve detailed optimization of the target of interest, such as Fig. 12(b). Although there are still some detection errors, HAFM can help to properly mitigate the misjudgments in the Baseline, such as Fig. 12(a), (c), and (d). These results generally indicate that HAFM plays an important role in BTAF, especially showing a significant effect enhancement on the WHU-CD dataset with higher spatial resolution. This also shows that HAFM has a good ability to deal with the spatial details of features. In order to verify the individual effects of different attention modules in HAFM and their joint effect, we also conducted experiments for verification, as shown in Table VI. No. 1 and No. 2 represent the results of using ECA or MSA alone in HAFM, and No. 3 is the result of using them together. From this result, it can be analyzed that ECA plays a more important role in HAFM than MSA, and HAFM achieves a more significant effect when they are used together. Furthermore, we conducted ablation experiments on the position embedding (PE) of HAFM to validate its effectiveness in screening out valuable information in the encoder and enhancing the features containing change information in the decoder.

2) Ablation on BTB: BTB divides change features into semantic tokens and SFVs and strengthens the model's context modeling ability and spatial awareness of interesting changes by branching multihead attention. No. 3 in Table V shows the quantitative results of BTB added to the baseline alone. Compared with the baseline of No. 1, the F1-score of No. 3 in the three datasets was 2.7%/4.49%/5.06% higher, and the IoU was 3.87%/8.50%/6.60% higher, respectively. According to Fig. 12, when BTB is added to the baseline alone, the model achieves more effective recognition and localization performance for changing targets, but it has a relatively rough target edge processing capability. When we combine BTB and HAFM at the same time, the model achieves an optimal detection effect. These results illustrate that BTB plays an important role in our model, which can help the model to reasonably and effectively understand the relationship between

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TABLE V
QUANTITATIVE RESULTS (%) OF ABLATION EXPERIMENTS WITH HAFM AND BTB ON THREE DATASETS, REPORTING RESULTS FOR F1-SOCRE AND IOU

No.	Method	LEVIR-CD		WHU-CD		GZ-CD	
		F1	IoU	F1	IoU	F1	IoU
1	Baseline	88.09	79.03	83.87	71.91	78.17	65.48
2	Baseline+HAFM	90.24	82.07	90.11	82.27	82.16	70.15
3	Baseline+BTB	90.79	82.90	88.36	80.41	83.23	72.08
4	BTAFNet	91.65	84.59	93.09	87.07	85.75	75.05

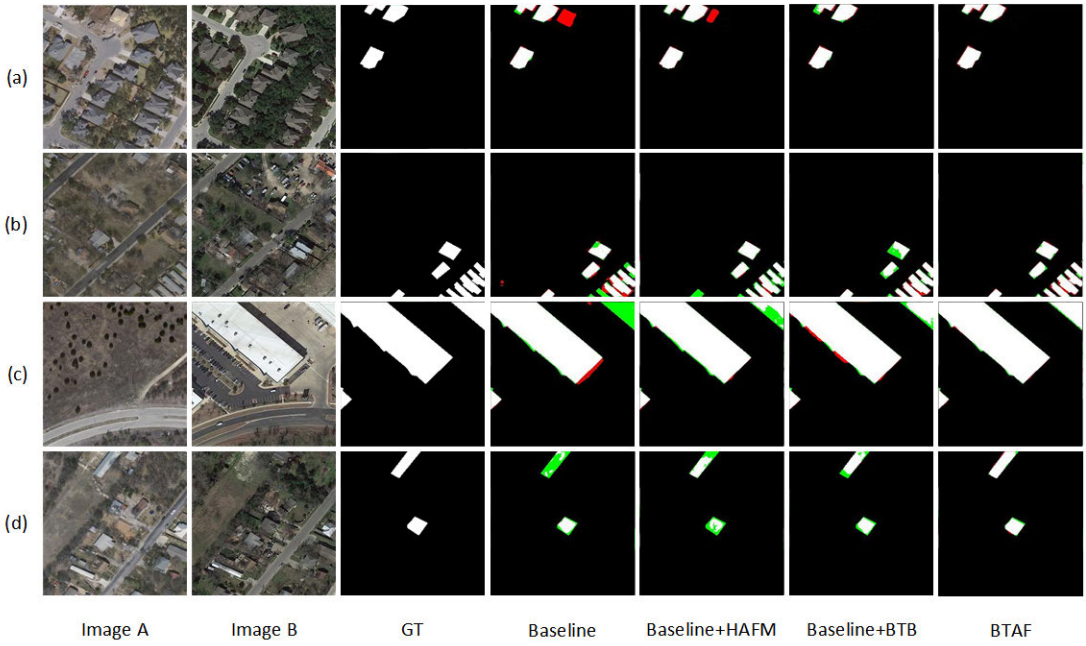


Fig. 12. Visualization results of ablation experiments for different modules in BTAFNet. (a)–(d) Visualization results of ablation experiments on different selected pictures, taking LEVIR-CD data as an example.

the change target of interest and its environment while paying more attention to the spatial details of the change features.

3) Ablation on Position Embedding: For CD tasks, both feature processing before the fusion of bitemporal RS images and the representation enhancement of change feature reflecting differences are necessary. Therefore, in BTAFNet, we jointly achieve both processes using a funnel-shaped filtering structure designed based on HAFM for multilevel features. We conducted ablation experiments on the PE of HAFM to validate its effectiveness in screening out valuable information in the encoder and enhancing the representation of the change feature in the decoder. We used the case without HAFM's PE as a baseline, and the experimental results are shown in Table VII. The results indicate that when we apply HAFM's PE separately in the encoder and decoder, our model consistently and significantly improves the F1-score and IoU on all three CD datasets. Furthermore, when both the encoder and decoder have PE, our method achieves optimal performance. These results indicate that filtering and enhancing multilevel features before and after computing differences between bitemporal images is both reasonable and effective. Therefore, we applied PE in both the encoder and decoder of BTAFNet.

E. Parameter Analysis

In this section, we will conduct analytical experiments on the width of branch self-attention and the role of the SFV in BTB on three different datasets to validate that our model has achieved a local optimum in the parameter design of this module. The width of multihead self-attention (S.W) is an important influencing factor for BTB, as it effectively enhances the model's self-attention learning capability. The different self-attention branches determined by S.W combine with the SFV branch to integrate context and spatial detail information in the change feature. We also conducted an additional analysis of the impact of introducing SFV under different S.W configurations. Our analytical experiments involved running S.W from 1 to 3, accompanied by the introduction or absence of SFV. The experiments were carried out sequentially, primarily testing the change in the F1-score and IoU of our network under different parameter configurations. The results of the analytical experiments are presented in Table VIII. Specifically, when the model only includes a single-self-attention branch and multibranch input (No. 1), it obtained 91.23%/92.80%/85.13% F1-score and 83.73%/86.44%/74.18% IoU on the three datasets. This

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TABLE VI

IMPACT OF DIFFERENT ATTENTION MECHANISMS IN HAFM, INCLUDING THE ABLATION OF ECA AND MSA. THE RESULTS (%) OF ABLATION EXPERIMENTS ON DIFFERENT DATASETS ARE REPORTED

No.	Attention	LEVIR-CD		WHU-CD		GZ-CD	
		F1	IoU	F1	IoU	F1	IoU
1	ECA	91.43	84.02	92.51	85.98	85.60	74.29
2	MSA	91.24	83.88	92.05	85.46	85.37	73.81
3	HAFM	91.65	84.59	93.09	87.07	85.75	75.05

TABLE VII

ABLATION EXPERIMENTS ON PE OF THE HAFM ON THREE DIFFERENT CD DATASETS. F1-SCORE AND IOU WERE REPORTED SEPARATELY FOR THE PE IN THE ENCODER (PE.E) AND THE PE IN THE DECODER (PE.D)

No.	PE.E	PE.D	LEVIR-CD		WHU-CD		GZ-CD	
			F1	IoU	F1	IoU	F1	IoU
1	×	×	90.79	82.90	88.36	80.41	83.23	72.08
2	✓	×	91.46	84.26	92.07	85.30	84.82	73.98
3	×	✓	91.36	84.09	92.04	85.27	84.42	73.04
4	✓	✓	91.65	84.59	93.09	87.07	85.75	75.05

TABLE VIII

INFLUENCE OF DIFFERENT SELF-ATTENTION WIDTHS (S.W) IN BMSA AND WHETHER TO INPUT SFV ON THE MODEL. THESE WIDTHS INCLUDE 1, 2, 3, T, AND F, RESPECTIVELY, INDICATE WHETHER BMSA IS ACCOMPANIED BY THE INPUT OF SFV. QUANTITATIVE RESULTS (%) ON DIFFERENT DATASETS ARE REPORTED

No.	S.W	With SFV	LEVIR-CD		WHU-CD		GZ-CD	
			F1	IoU	F1	IoU	F1	IoU
1	1	T	91.23	83.73	92.80	86.44	85.13	74.18
2	2(ours)	T	91.65	84.59	93.09	87.07	85.75	75.05
3	3	T	91.31	84.08	92.82	86.77	85.34	74.46
4	1	F	90.96	83.24	92.59	85.93	85.09	74.28
5	2	F	91.37	83.96	92.83	86.45	85.37	74.70
6	3	F	91.18	83.63	92.64	86.08	85.16	74.22

result is higher than the single-self-attention branch and the tokens-only input branch model (No. 4), which shows that the combination of tokens and SFV self-attention is effective. When we increase the number of self-attention branches to 2 and have the SFV, the model achieves an F1-score of 91.65%/93.09%/85.75% and an IoU of 84.59%/87.07%/75.05%, which achieves optimal performance. This shows that improving the self-attention branch and additionally introducing the SFV can effectively improve detection accuracy. However, when the number of self-attention branches increases to 3 (No. 3 and No. 6), the indicator drops instead. This may be because too many self-attention branches make the information fused too complex and affect reasonable self-attention to change information. Therefore, in the end, we chose a self-attention branch count of 2, along with the input of SFV, as the final configuration for BTB.

F. Network Visualization

In order to provide a more intuitive understanding of the functions of different modules in our model, we offer visualizations of feature maps at various stages of the BTAF model. The visualized results are based on a pair of bitemporal images selected from the LEVIR-CD test set as model input. For the given input images [Fig. 13(a)], they are first fed into the multiscale residual network backbone to obtain features at different levels. These features are then filtered through HAFM to learn and effectively fuse feature dependencies among multilevel features, resulting in refined features F_{pre1}, F_{pre2}, F_{pre3}, F_{pos1}, F_{pos2}, and F_{pos3} [#1, #2, and #3 in Fig. 13(b)]. From Fig. 13(a)–(e), we can observe that higher level features gradually exhibit clearer contours and shapes after continuously combining lower level features through attention fusion while ensuring accurate positioning of the target. Afterward, these refined bitemporal features are passed to the decoder to compute feature differences and are once again filtered through HAFM to obtain a representative change feature F_c [Fig. 13(c)]. F_c is then divided into two branches, namely, tokens obtained from the embedding layer and the spatial vector obtained from ESA. These branches are input to BTB to obtain F_b [Fig. 13(d)]. We can observe that F_b exhibits excellent spatial details and accurately distinguishes the changing target. Finally, a convolution-based prediction head generates the change probability map [Fig. 13(e)]. These results demonstrate the effectiveness of the proposed model.

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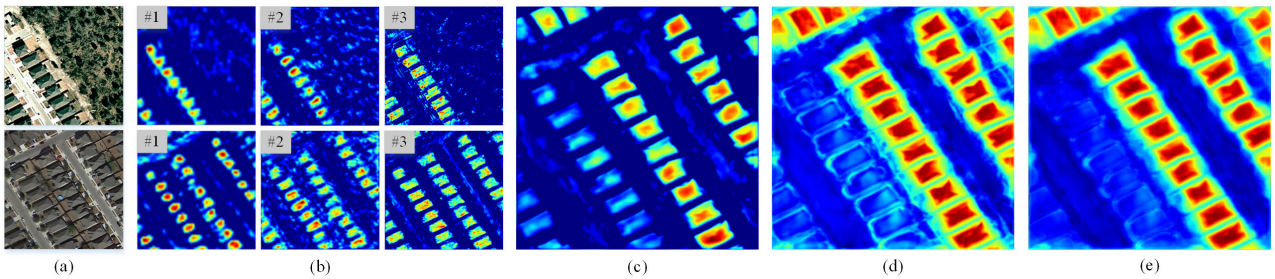


Fig. 13. Example of network visualization using a pair of bitemporal images selected from the LEVIR-CD test set as input. (a) Pair of input bitemporal images. (b) Multiscale refined bitemporal feature maps obtained by HAFM F_{pre1}, F_{pre2}, F_{pre3}, F_{pos1}, F_{pos2}, and F_{pos3}. (c) Refined change feature mapF_c. (d) Feature map F_b after BTB. (e) Change probability map.

V. Conclusion

In this article, we propose a hybrid attention fusion and branch transformer-based network for change detection in HR remote sensing images. The network utilizes multilevel residual blocks to capture multiscale features and employs the proposed HAFM to perform multilevel blending and differentiation across spatial and channel dimensions, yielding a finely detailed change feature. This process adopts a Siamese architecture to improve the model's efficiency. Furthermore, we propose a BTB to efficiently aggregate change information. Compared with the self-attention in traditional transformers, this module further combines a CNN branch to enhance the feature representation of local information and adopts a parallel branch structure to guide the learning of self-attention matrices with different receptive fields. We conducted extensive qualitative and quantitative experiments on various HR remote-sensing datasets, and the results demonstrate the effectiveness of our method. On the other hand, our method leads to higher computational resource consumption due to the use of low-granularity features in the attention-filtering process. Therefore, our future work will focus on an in-depth study of lightweight models. Our goals also include the more efficient application of branched transformer blocks to a broader range of usable tasks, such as semantic segmentation and medical image segmentation.

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Yu Shangguan received the bachelor's degree from the School of Mathematics and Computer Science, Yan'an University, Yan'an, Shaanxi, China, in 2020. He is currently pursuing the master's degree with the School of Computer Science and Technology, Yantai Technology and Business University, Yantai, Shandong, China.

His research interests include computer graphics (CG), computer vision, and image processing.



Jinjiang Li received the B.S. and M.S. degrees in computer science from the Taiyuan University of Technology, Taiyuan, China, in 2001 and 2004, respectively, and the Ph.D. degree in computer science from Shandong University, Jinan, China, in 2010.

From 2004 to 2006, he was an Assistant Research Fellow with the Institute of Computer Science and Technology, Peking University, Beijing, China. From 2012 to 2014, he was a Post-Doctoral Fellow with Tsinghua University, Beijing. He is currently a Professor with the School of Computer Science and Technology, Shandong Technology and Business University, Yantai, China. His research interests include image processing, computer graphics (CG), computer vision, and machine learning.



Yepeng Liu received the bachelor's and Ph.D. degrees from the School of Computer Science and Technology, Shandong University, Jinan, China, in 2014 and 2020, respectively.

He is currently a Teacher with Shandong Technology and Business University, Yantai, China. His research interests include image super-resolution, image denoising, and medical image processing.



Fan Zhang received the B.S. and Ph.D. degrees in computer science from Shandong University, Jinan, China, in 2009 and 2015, respectively.

From 2012 to 2014, he was a Joint-Training Ph.D. Student with the Department of Computer Science, University of Kentucky, Lexington, KY, USA. He is currently an Associate Professor with the School of Computer Science and Technology, Shandong Technology and Business University, Yantai, China. His research interests include image processing, computer vision, computer graphics (CG), and computer aided geometric design (CAGD).



Caiming Zhang received the B.S. and M.E. degrees in computer science from Shandong University, Jinan, China, in 1982 and 1984, respectively, and the Ph.D. degree in computer science from the Tokyo Institute of Technology, Tokyo, Japan, in 1994.

From 1997 to 2000, he held a visiting position at the University of Kentucky, Lexington, KY, USA. He is currently a Professor and a Ph.D. Supervisor with the Software College, Shandong University. His research interests include computer aided geometric design (CAGD), computer graphics (CG), image processing, and computational finance.

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